Basic Network Packet Logger using Python

Introduction

With the increasing use of the internet, monitoring network traffic has become essential for maintaining security and understanding data flow. Packet sniffing is a technique used to capture and analyze network packets transmitted over a network. This project focuses on developing a basic network packet logger using Python to capture and display source and destination IP addresses along with protocols in real time.

Abstract

The Basic Network Packet Logger is a Python-based tool that captures live network traffic using the Scapy library. It extracts key information such as source IP address, destination IP address, and protocol type (TCP/UDP) from each packet. The system displays this information in real time and identifies external IP addresses to highlight potential suspicious activity. This project demonstrates fundamental concepts of network monitoring and packet analysis.

Tools Used

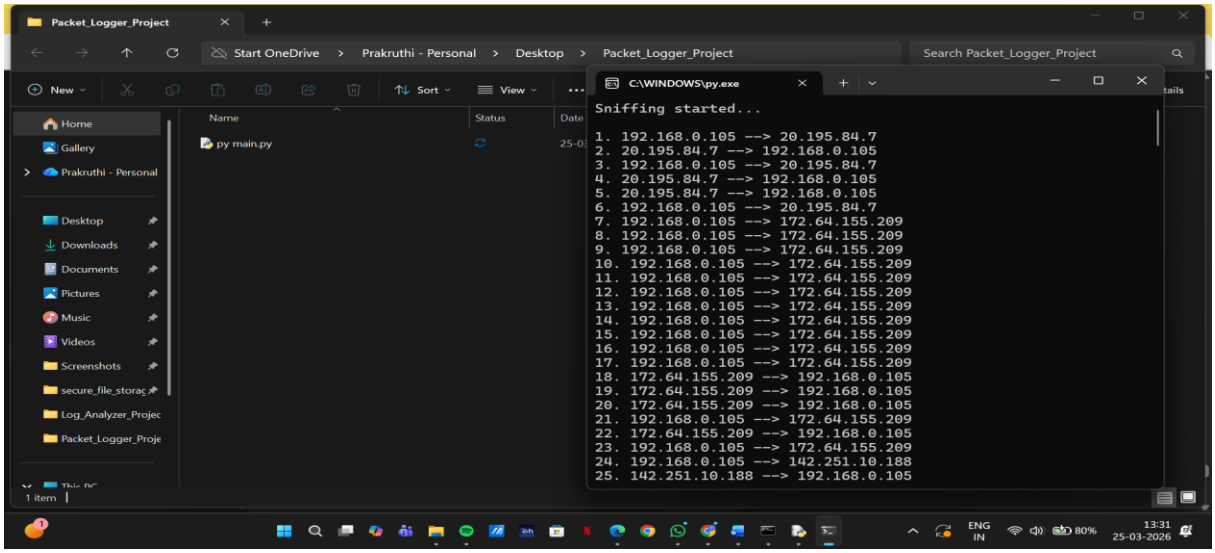
- Python
- Scapy Library
- Command Prompt (Administrator Mode)

Steps Involved in Building the Project

1. Installed Python and required libraries such as Scapy.
2. Created a Python script to capture network packets.
3. Used Scapy's sniffing function to monitor live traffic.
4. Extracted IP layer information including source and destination addresses.
5. Identified protocol types such as TCP and UDP.
6. Displayed packet details in real time.
7. Added a feature to detect and highlight external IP addresses.

Conclusion

The project successfully demonstrates how network packets can be captured and analyzed using Python. By displaying real-time traffic and identifying external IP addresses, the system provides basic insights into network activity. This tool can be further enhanced with features such as logging, alerts, and graphical interfaces to improve usability and security monitoring.



```
C:\WINDOWS\py.exe

Sniffing started...

1. 192.168.0.105 --> 40.104.68.50 | TCP
2. 40.104.68.50 --> 192.168.0.105 | TCP
3. 192.168.0.101 --> 255.255.255.255 | UDP
4. 192.168.0.105 --> 20.195.84.7 | TCP
5. 20.195.84.7 --> 192.168.0.105 | TCP
6. 3.111.241.224 --> 192.168.0.105 | TCP
7. 192.168.0.105 --> 3.111.241.224 | TCP
8. 192.168.0.105 --> 142.251.156.119 | UDP
9. 142.251.156.119 --> 192.168.0.105 | UDP
10. 192.168.0.105 --> 239.255.255.250 | UDP
11. 192.168.0.105 --> 20.195.84.7 | TCP
12. 20.195.84.7 --> 192.168.0.105 | TCP
13. 192.168.0.105 --> 192.168.0.1 | UDP
14. 192.168.0.105 --> 192.168.0.1 | UDP
15. 192.168.0.1 --> 192.168.0.105 | UDP
16. 192.168.0.1 --> 192.168.0.105 | UDP
17. 192.168.0.105 --> 35.186.224.24 | UDP
18. 192.168.0.105 --> 35.186.224.24 | UDP
19. 192.168.0.105 --> 35.186.224.24 | TCP
20. 35.186.224.24 --> 192.168.0.105 | UDP
21. 35.186.224.24 --> 192.168.0.105 | UDP
22. 35.186.224.24 --> 192.168.0.105 | UDP
23. 192.168.0.105 --> 35.186.224.24 | UDP
24. 192.168.0.105 --> 35.186.224.24 | UDP
25. 192.168.0.105 --> 35.186.224.24 | UDP
26. 35.186.224.24 --> 192.168.0.105 | TCP
27. 192.168.0.105 --> 35.186.224.24 | TCP
28. 192.168.0.105 --> 35.186.224.24 | TCP
29. 192.168.0.105 --> 35.186.224.24 | TCP
30. 35.186.224.24 --> 192.168.0.105 | UDP
```

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C:\WINDOWS\py.exe

Sniffing started...

1. 192.168.0.105 --> 20.195.84.7
2. 20.195.84.7 --> 192.168.0.105
3. 192.168.0.105 --> 20.195.84.7
4. 20.195.84.7 --> 192.168.0.105
5. 20.195.84.7 --> 192.168.0.105
6. 192.168.0.105 --> 20.195.84.7
7. 192.168.0.105 --> 172.64.155.209
8. 192.168.0.105 --> 172.64.155.209
9. 192.168.0.105 --> 172.64.155.209
10. 192.168.0.105 --> 172.64.155.209
11. 192.168.0.105 --> 172.64.155.209
12. 192.168.0.105 --> 172.64.155.209
13. 192.168.0.105 --> 172.64.155.209
14. 192.168.0.105 --> 172.64.155.209
15. 192.168.0.105 --> 172.64.155.209
16. 192.168.0.105 --> 172.64.155.209
17. 192.168.0.105 --> 172.64.155.209
18. 172.64.155.209 --> 192.168.0.105
19. 172.64.155.209 --> 192.168.0.105
20. 172.64.155.209 --> 192.168.0.105
21. 192.168.0.105 --> 172.64.155.209
22. 172.64.155.209 --> 192.168.0.105
23. 192.168.0.105 --> 172.64.155.209
24. 192.168.0.105 --> 142.251.10.188
25. 142.251.10.188 --> 192.168.0.105
26. 192.168.0.105 --> 239.255.255.250
27. 192.168.0.105 --> 192.168.0.1
28. 192.168.0.105 --> 192.168.0.1
29. 192.168.0.1 --> 192.168.0.105
30. 192.168.0.1 --> 192.168.0.105
```

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C:\WINDOWS\py.exe

Sniffing started...

1. 192.168.0.105 --> 142.251.10.188 | TCP
2. 142.251.10.188 --> 192.168.0.105 | TCP
⚠ External IP detected: 142.251.10.188
3. 192.168.0.105 --> 142.251.222.196 | UDP
4. 142.251.222.196 --> 192.168.0.105 | UDP
⚠ External IP detected: 142.251.222.196
5. 192.168.0.105 --> 192.168.0.1 | UDP
6. 192.168.0.105 --> 192.168.0.1 | UDP
7. 192.168.0.1 --> 192.168.0.105 | UDP
8. 192.168.0.1 --> 192.168.0.105 | UDP
9. 192.168.0.105 --> 35.186.224.24 | UDP
10. 192.168.0.105 --> 35.186.224.24 | UDP
11. 35.186.224.24 --> 192.168.0.105 | UDP
⚠ External IP detected: 35.186.224.24
12. 35.186.224.24 --> 192.168.0.105 | UDP
⚠ External IP detected: 35.186.224.24
13. 35.186.224.24 --> 192.168.0.105 | UDP
⚠ External IP detected: 35.186.224.24
14. 35.186.224.24 --> 192.168.0.105 | UDP
⚠ External IP detected: 35.186.224.24
15. 35.186.224.24 --> 192.168.0.105 | UDP
⚠ External IP detected: 35.186.224.24
16. 192.168.0.105 --> 35.186.224.24 | UDP
17. 192.168.0.105 --> 35.186.224.24 | UDP
18. 192.168.0.105 --> 35.186.224.24 | UDP
19. 192.168.0.105 --> 104.18.32.47 | UDP
20. 192.168.0.105 --> 104.18.32.47 | UDP
21. 192.168.0.105 --> 104.18.32.47 | UDP
22. 192.168.0.105 --> 104.18.32.47 | UDP
23. 192.168.0.105 --> 104.18.32.47 | UDP
24. 192.168.0.105 --> 104.18.32.47 | UDP
```